RESEARCH PAPER • FIELD DYNAMICS • PAPER III

Distributed Field Processing III: The Unit of Gravity

Deriving the Tile Size at Electron Scale

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ABSTRACT

The Question: Papers I and II established that gravity operates through distributed field closure with $k = 4$ tetrahedral topology. But topology is scale-free: it gives a shape, not a size. What sets the tile size of the gravitational field?

The Answer: Two independent routes converge on the same scale. Route 1 (electromagnetic stability) gives $r_e = e^2/(4\pi\varepsilon_0 m_e c^2) = 2.818$ fm. Route 2 (gravitational topology) gives $r_e = k\sqrt{Gm_e/a_0} = 2.82$ fm. These routes use completely different physics: one contains no gravity, the other contains no electromagnetism. Yet they converge to 0.21%. This convergence is unique to the electron. The unit of gravity is the electron-scale tile.

The Topological Ratio: The convergence reveals a deeper structure. Two independent derivations of a_0 (cosmological and topological) differ by exactly $\sqrt{k/\pi} = 1.128$ to 0.1%. This ratio connects flat-space MOND to curved cosmological spacetime.

The Prediction: From the MOND equation $a_0 = cH/(2\pi)$, this paper predicts $H = 78.90$ km/s/Mpc. Paper IV subsequently proved that this value is H_{flat} (what MOND predicts in flat space), and that the topological correction yields $H_{\text{measured}} = 69.92$ km/s/Mpc, matching the GW170817 gravitational wave measurement to 0.11%.

Gravitational Tile | Electron Scale | MOND | Tetrad Topology | Classical Electron Radius

Field Closure

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1. Introduction

Papers I and II established that gravity operates through distributed field closure with $k = 4$ tetrahedral topology. But topology alone cannot determine a physical scale. This paper asks: what sets the tile SIZE? The answer connects gravitational topology to the electron scale through a remarkable convergence of independent physics.

1.1 The Question

Gravity has topology. The number $k = 4$ appears throughout gravitational physics:

- **Simplex theorem:** The minimum 3D enclosure has $d + 1 = 4$ faces
- **ADM formalism:** 1 Hamiltonian + 3 momentum constraints
- **Tetrad formalism:** 1 timelike + 3 spacelike basis vectors
- **LIGO polarizations:** $10 - 2k = 2$ independent modes

Papers I and II derived this topology from first principles. But $k = 4$ is a pure number. It tells us the gravitational tile is tetrahedral. It does not tell us how big that tetrahedron is. A tetrahedron can be atomic-scale or cosmological-scale; the topology is identical.

> **The Central Question**
>
> **Topology gives $k = 4$. What gives the tile SIZE?**
>
> This paper answers that question. Two independent routes, using completely different physics, converge on the same scale: $r_e = 2.82$ fm.

Why understanding the mechanism matters. Consider an analogy. A GPU returns correct computational results. Knowing this is sufficient for using the GPU. But understanding HOW it achieves those results reveals architecture: thousands of local threads processing in parallel, each handling its piece independently. This architectural insight enables optimization, extension, and prediction in ways that "it works" cannot.

The same principle applies here. Knowing THAT the topology is $k = 4$ suffices for many calculations. But understanding HOW this topology determines a specific scale reveals deeper

structure: the crossover between gravitational topology and electromagnetic stability at the electron scale. This is what connects the tile size to testable predictions about the Hubble constant.

1.2 Where This Paper Fits

This is Paper III in the Distributed Field Processing series. Each paper establishes a distinct result; together they form a complete framework:

Paper	Question	Key Result
I	How does gravity couple locally?	$\mu(x) = x/(1+x)$, galactic dynamics
II	What sets the acceleration scale?	$a_0 = cH/(2\pi)$, cosmological connection
III (this)	**What sets the tile size?**	$r_e = 2.82$ **fm, electron-scale tile**
IV	What does topology predict for H?	$H_{\text{measured}} = 69.92$ km/s/Mpc
V	How does measurement work?	Measurement cage, bidirectional derivation

Table 1: The DFP paper series. Each paper answers a specific question; the results build on each other.

The logical chain: Paper I derives the coupling function. Paper II derives the acceleration scale. This paper (III) derives the tile size. Paper IV applies the topology to predict H_{measured}. Each step requires the previous ones.

1.3 Scope and Commitments

Before proceeding, we state explicitly what this paper establishes and what it does not claim.

Scope and Commitments

What this paper ESTABLISHES:

- Tile size $r_e = 2.82$ fm from two independent routes (0.21% convergence)
- The ratio $\sqrt{k/\pi} = 1.128$ connecting topology to cosmology
- Testable prediction: $H = 78.90$ km/s/Mpc from the MOND equation (§4.3)

What this paper does NOT claim:

- We do not derive r_e from pure topology (scale emerges from convergence)
- We do not prove MOND is "correct" (we explain why a_0 has its value)

Subsequent development (Paper IV):

- Paper IV proved: MOND + $k = 4$ topology distinguishes H_{flat} from H_{measured}
- The $H = 78.90$ prediction is H_{flat}; topology yields $H_{\text{measured}} = 69.92$
- See §4.4 for the evolution of understanding

2. From Shape to Size

Section 1 established the question: topology gives $k = 4$, but what gives the tile SIZE? This section builds the bridge. We have two ingredients, each powerful but incomplete. Connecting them requires understanding how gravity couples through the tetrad structure, which leads directly to the crossover condition that Section 3 will use.

2.1 Two Ingredients, Neither Complete

We have two pieces of physics that should determine the gravitational tile. Neither, on its own, can give us a length.

Ingredient 1: Topology gives a pure number. The simplex theorem states that the minimum convex polytope enclosing volume in d dimensions has $d + 1$ faces. For 3D space: $k = d + 1 = 4$. This is the tetrahedron, the simplest possible closed volume.

The number $k = 4$ is dimensionless. It appears at every scale, from molecular (CH_4, SiO_4) to cosmological (Regge calculus tiles spacetime with 4-simplices). It tells us the gravitational tile is tetrahedral. It cannot tell us how big that tetrahedron is.

Ingredient 2: MOND gives an acceleration, not a length. Paper II derived $a_0 = cH/(2\pi)$ from horizon thermodynamics: the Unruh temperature at acceleration a_0 equals the Gibbons-Hawking

temperature of the cosmological horizon. This gives a characteristic acceleration:

$$a_0 = \frac{cH}{2\pi} \approx 1.2 \times 10^{-10}\ \mathrm{m/s^2}$$

But acceleration has units of m/s^2, not meters. The MOND scale tells us where the crossover between Newtonian and non-Newtonian regimes occurs. It does not specify which particle or which radius sets the tile size.

The gap. We have shape ($k = 4$) and we have an acceleration scale (a_0). To get a length, we need to connect them. The connection comes from understanding how gravity actually couples through the $k = 4$ topology.

2.2 How Gravity Couples: The Tetrad Structure

The $k = 4$ topology is not merely a counting exercise. It has physical consequences. To see this, we use the tetrad formalism, which is standard graduate-level general relativity (Misner, Thorne & Wheeler; Wald).

Why tetrads? The equivalence principle requires physics to be locally special-relativistic at every point in curved spacetime. The tetrad (or vierbein) formalism makes this explicit. At each point, we define four orthonormal basis vectors e^a_μ that span the local tangent space. The metric is then:

$$g_{\mu\nu} = \eta_{ab}\, e^a_\mu\, e^b_\nu$$

Here $\eta_{ab} = \mathrm{diag}(-1, +1, +1, +1)$ is the flat Minkowski metric. The tetrad converts between the curved spacetime (indices μ, ν) and the local flat frame (indices a, b). This is the same physics as standard GR, expressed in different language:

Concept	GR Language	Tetrad Language
Field defined everywhere	Non-degenerate metric	Closure
Mathematical condition	$\det(g) \neq 0$	$\det(e) \neq 0$
Structure	3+1 decomposition	4 tetrad legs
Constraints	4 ADM equations	4 projections onto legs
Packing topology	*(not named)*	$k = 4$ faces

Table 1: GR and tetrad formalisms express the same physics. Both require fields to be complete in four directions.

Where does $k^2 = 16$ come from? The tetrad e^a_μ has two indices. The frame index a runs from 0 to 3 (four values). The coordinate index μ also runs from 0 to 3 (four values). This gives $4 \times 4 = 16$ independent components.

When the metric equation couples geometry to frame structure, it sums over both indices. The $k^2 = 16$ is not a fit parameter; it is the number of components in the tetrad field, which follows from the dimensionality of spacetime.

Isotropy engages all channels. Gravity is isotropic: no spatial direction is preferred. In tetrad language, this means no frame direction e^a is privileged. Full isotropic coupling therefore engages all $k^2 = 16$ tetrad components simultaneously.

For a mass m at radius r, the topologically-coupled gravitational acceleration is the Newtonian value multiplied by the coupling factor:

$$a_{\text{coupled}} = k^2 \cdot \frac{Gm}{r^2} = 16 \cdot \frac{Gm}{r^2}$$

This is the key result: the $k = 4$ topology amplifies the gravitational coupling by a factor of $k^2 = 16$. Gravity couples through all 16 constraint channels, one for each combination of tetrad legs.

2.3 The Crossover Condition

Now we can connect the two ingredients. Where does the topologically-coupled gravitational acceleration equal the MOND scale? This crossover defines the characteristic tile size.

Setting the coupled acceleration equal to a_0:

$$k^2 \cdot \frac{Gm}{r^2} = a_0$$

Solving for r:

$$r^2 = \frac{k^2\, Gm}{a_0} \quad \Rightarrow \quad r = k \cdot \sqrt{\frac{Gm}{a_0}}$$

> **The Crossover Condition**
>
> $$r = k \cdot \sqrt{\frac{Gm}{a_0}}$$
>
> This defines a characteristic length scale for any mass m. The factor k appears because k^2 enters the acceleration, and taking the square root extracts one power of k.

Why this works. The crossover condition takes the two scale-free ingredients and combines them in the only way that produces a length:

- Topology contributes $k = 4$ (dimensionless)
- MOND contributes a_0 (units: m/s^2)
- Gravity contributes G and m (units: m^3/kg·s^2 and kg)
- The combination $\sqrt{Gm/a_0}$ has units of meters

Why the electron? The crossover condition defines a length for any mass m. Which mass matters? The electron is the smallest stable particle that satisfies the crossover. It is a probe, not a cause, like a thermometer that reveals temperature but does not create it. We measure through the electron because it inhabits the scale where topology meets cosmology.

> **Section 2 Summary**
>
> - **Topology** gives $k = 4$ (dimensionless, no length)
> - **MOND** gives a_0 (acceleration, no length)
> - **Tetrad coupling** gives $k^2 = 16$ (from 4×4 components)
> - **Crossover condition:** $r = k\sqrt{Gm/a_0}$
>
> **Next:** Section 3 derives r_e from two independent routes and shows they converge to 0.21%.

3. Two Routes to the Tile Size

Section 2 established the crossover condition: $r = k\sqrt{Gm/a_0}$. This section applies it. We derive the tile size from two completely independent domains of physics that should have nothing to do with each other. The fact that they converge to 0.21% is the central result of this paper.

3.1 Route 1: Electromagnetic Stability

The first route comes from pure electromagnetism. It asks: at what scale does the electron's electromagnetic field energy equal its rest mass? This question involves no gravity, no cosmology, no topology.

The physical picture. A charged particle creates an electric field that extends outward. This field contains energy. The closer you get to the charge, the stronger the field, and the more energy is stored. At some radius, the electromagnetic energy stored in the field equals the particle's rest mass energy.

Electromagnetic self-energy of a charged sphere of radius r:

$$U_{\mathrm{EM}} = \frac{e^2}{4\pi\varepsilon_0 r}$$

Rest mass energy of the electron:

$$E = m_e c^2$$

Setting $U_{\text{EM}} = E$ and solving for r:

Route 1 Result:

$$r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} = 2.818 \times 10^{-15}\ \text{m} = 2.818\ \text{fm}$$

Constants used: e, ε_0, m_e, c
Domain: Electromagnetism + special relativity
No gravity. No cosmology. No topology.

This is the classical electron radius, a textbook result dating to Lorentz. It defines the scale where electromagnetic self-interaction becomes significant.

3.2 Route 2: Gravitational Topology

The second route comes from the crossover condition derived in §2.3. It asks: at what scale does topologically-coupled gravitational acceleration equal the MOND scale? This question involves no electromagnetism.

The crossover condition (from §2.3):

$$r = k \cdot \sqrt{\frac{Gm}{a_0}}$$

Applying to the electron with measured values:

- $k = 4$ (tetrahedral closure, from simplex theorem)
- $m_e = 9.109 \times 10^{-31}$ kg (electron mass)
- $a_0 = 1.22 \times 10^{-10}$ m/s² (MOND empirical scale)
- $G = 6.674 \times 10^{-11}$ m³/(kg·s²) (gravitational constant)

Calculation:

$$r_e = 4 \times \sqrt{\frac{(6.674 \times 10^{-11})(9.109 \times 10^{-31})}{1.22 \times 10^{-10}}}$$

Route 2 Result:

$$r_e = k \cdot \sqrt{\frac{G m_e}{a_0}} = 2.82 \times 10^{-15}\ \text{m} = 2.82\ \text{fm}$$

Constants used: G, k, m_e, a_0
Domain: Gravity + cosmology + topology
No electromagnetism.

3.3 The Convergence

Compare the two results. Route 1 uses electromagnetism and special relativity. Route 2 uses gravity, cosmology, and topology. They share only one constant: the electron mass m_e. Yet they agree to four significant figures:

Route	Constants Used	Result	Domain
1 (EM)	e, ε_0, m_e, c	2.818 fm	Electromagnetism
2 (Grav)	G, k, m_e, a_0	2.82 fm	Gravity + Cosmology

Route	Constants Used	Result	Domain
Convergence		**0.21%**	

Table 2: Two routes to the tile size. Different physics, same scale.

Why this convergence is remarkable. The two routes use different physics. They share no constants except m_e. Route 1 knows nothing about gravity or the cosmological horizon. Route 2 knows nothing about electric charge or the permittivity of free space. Yet they land on the same scale.

Why this convergence is unique. The agreement is specific to the electron. For the muon ($m_\mu = 207 m_e$), the two routes disagree by a factor of ~10^3. For the proton ($m_p = 1836 m_e$), they disagree by ~10^7. Only for the electron do electromagnetic stability and gravitational topology meet at the same scale.

> **The Central Result**
>
> Two independent physics domains converge on $r_e = 2.82$ fm to 0.21%. This is the unit of gravity: the tile size of the gravitational field.

What the convergence enables. Because the two routes agree, we can derive a_0 from first principles. The derivation chain runs from pure topology to the MOND acceleration scale:

Step	Result	Source
1	$k = 4$	3D simplex theorem
2	Tile: r_e, m_e	Convergence (Routes 1 & 2)
3	$g_{\text{tile}} = G m_e / r_e^2$	Gravitational coupling at tile scale
4	4 coupling faces	Tetrahedral closure
5	$k^2 = 16$	Isotropic coupling (all 16 channels)
6	$a_0 = 16 G m_e / r_e^2$	Total coupling = MOND scale

Table 3: The derivation chain from topology to a_0.

What this proves. The coefficient 16 is not a fit parameter. It is $k^2 = (d+1)^2 = 4^2$, where $d = 3$ is the dimension of space. The only empirical inputs are measured values of G, m_e, and r_e. The formula itself follows from the tetrad structure.

3.4 Paper IV Confirmation

Paper IV provides a third independent route using the tetrad formalism. When Einstein's equations are expressed in tetrad form, they become a hyperbolic system with characteristic speed c. Field constraints propagate at the speed of light; we call this the "closure rate."

The cosmological horizon lies at distance $R_H = c/H$, set by how far constraint information can propagate in one Hubble time. Combining the closure rate c with the cosmological boundary H gives:

$$r_e = k \cdot \sqrt{\frac{2\pi G m_e}{cH}} = 2.82 \text{ fm}$$

Three routes. Three different combinations of physics. Same result. The tile size is not assumed; it is derived.

> **Section 3 Summary**
>
> - **Route 1 (EM):** $r_e = 2.818$ fm (no gravity)
> - **Route 2 (Grav):** $r_e = 2.82$ fm (no EM)
> - **Convergence:** 0.21% (unique to electron)
> - **Route 3 (Paper IV):** Tetrad formalism confirms
>
> **Next:** Section 4 explores the topological ratio $\sqrt{k/\pi}$ that connects the tile size to cosmological predictions.

4. The Topological Connection

The tile size derivation reveals a deeper topological structure: the ratio $\sqrt{k/\pi}$ that connects flat-space MOND to cosmological observables. This section explores why two a_0 derivations differ by this ratio, what it means physically, and how Paper IV applied it to predict H_{measured}.

4.1 The Two a_0 Values

There are two independent routes to the MOND acceleration scale:

Cosmological route (horizon thermodynamics):

$$a_0^{\text{cosmo}} = \frac{cH}{2\pi} = 1.08 \times 10^{-10}\ \text{m/s}^2$$

Topological route (tile size derivation):

$$a_0^{\text{geom}} = k^2 \cdot \frac{Gm_e}{r_e^2} = 1.22 \times 10^{-10}\ \text{m/s}^2$$

These differ by 12%. But look at the ratio:

$$\frac{a_0^{\text{geom}}}{a_0^{\text{cosmo}}} = \frac{1.22}{1.08} = 1.130$$

4.2 The √(k/π) Ratio

The observed ratio matches a simple topological expression:

$$\sqrt{\frac{k}{\pi}} = \sqrt{\frac{4}{\pi}} = \sqrt{1.273} = 1.128$$

Match: 0.1%

This is not coincidence. The ratio has physical meaning:

- $k = 4$: Tetrahedral closure (flat, discrete, local)
- π: Spherical horizon (curved, continuous, global)
- $\sqrt{k/\pi}$: Frame correction between these geometries

Why the square root? The Hubble constant $H = v/d$ has units of inverse time, which depends on length. The ratio k/π compares topological characteristics (area-like quantities). Since area scales as length squared, we take the square root to convert to a length correction.

> **The √(k/π) Ratio**
>
> $\sqrt{k/\pi} = 1.128$ is the frame correction between flat local frames (characterized by $k = 4$ tetrahedral closure) and the curved cosmological boundary (characterized by π from the spherical horizon).
>
> MOND operates in flat space—it was derived from galactic dynamics, where spacetime curvature is negligible. Cosmological observations, however, project through curved spacetime to reach the spherical horizon. The ratio connects these two regimes.

4.3 Testable Prediction: The Hubble Constant

The $\sqrt{k/\pi}$ ratio connects the tile size derivation to a testable cosmological prediction. From the MOND equation $a_0 = cH/(2\pi)$, we can predict the Hubble constant:

Paper III Prediction

According to the MOND equation:

$$H = \frac{2\pi a_0}{c} = \frac{2\pi \times (1.22 \times 10^{-10})}{3 \times 10^8} = 78.90 \text{ km/s/Mpc}$$

This is a testable prediction. The MOND acceleration scale a_0, derived from galactic dynamics, predicts a specific value for the Hubble constant.

The observational puzzle: A century of observations measures $H \approx 70$ km/s/Mpc, not 79. This 12% discrepancy requires explanation.

The resolution: The $\sqrt{k/\pi} = 1.128$ ratio—identified in this paper from the convergence of the two a_0 derivations—provides the explanation. This is documented in the Historical Note below.

4.4 Historical Note: Evolution of Understanding

Historical Note: Evolution of the Hubble Prediction

Original (Paper III v1-v3): The unified formula predicts $H = 78.90$ km/s/Mpc. When this paper was first written, we interpreted this as MOND's prediction for the Hubble constant—a testable value that could be compared directly to observation.

What Paper IV established: Paper IV ("Resolving the Hubble Tension Through Tetrad Topology") proved that when MOND is projected through $k = 4$ tetrad topology, a distinction emerges between H_{flat} and H_{measured}. The 78.90 value is H_{flat}—what MOND predicts in flat space. Cosmological observations project through the $k = 4$ field topology:

$$H_{\text{measured}} = \frac{H_{\text{flat}}}{\sqrt{k/\pi}} = \frac{78.90}{1.128} = 69.92 \textbf{ km/s/Mpc}$$

What this means: Both values are valid predictions for different physical situations. MOND in flat space predicts $H_{\text{flat}} = 78.90$ km/s/Mpc. MOND projected through $k = 4$ topology predicts $H_{\text{measured}} = 69.92$ km/s/Mpc. Paper IV established that cosmological observations measure through the topology, so H_{measured} is what observers see. This is a DFP contribution: the framework yields two distinct predictions, one for flat space and one for measurement through topology.

4.5 DFP Resolution: Paper IV

Paper IV ("Resolving the Hubble Tension Through Tetrad Topology") proved that MOND projected through $k = 4$ tetrad topology yields the H_{flat} vs H_{measured} distinction:

Paper IV Result

$$H_{\text{measured}} = \frac{H_{\text{flat}}}{\sqrt{k/\pi}} = \frac{78.90}{1.128} = 69.92 \text{ km/s/Mpc}$$

GW170817 measurement: 70.0 ± 12 km/s/Mpc
Match: 0.11%

Four quantitative tests confirm $k = 4$:

Test	Prediction	Observed	Match
Coupling ratio	$k^2 = 16$	15.93	0.41%
Frame correction	$\sqrt{k/\pi} = 1.128$	1.130	0.1%
H_{measured}	69.92 km/s/Mpc	70.0 (GW170817)	0.11%
Bidirectional	$H \to m_e$ closes	CODATA match	0.10%

Table 2: Four quantitative tests from Paper IV. All confirm $k = 4$.

Section 4 Summary

- Two a_0 routes differ by 12%
- The ratio is $\sqrt{k/\pi} = 1.128$ (0.1% match)
- This connects flat-space MOND to cosmological observation
- Paper III predicted $H = 78.90$ km/s/Mpc from MOND
- Paper IV proved: MOND + $k = 4$ topology yields H_{flat} vs H_{measured}
- Paper IV result: $H_{\text{measured}} = 69.92$ km/s/Mpc (0.11% match to GW170817)

5. Scientific Status

A framework is only as good as its tests. This section presents the quantitative predictions, specifies what would falsify the framework, and draws a clear boundary around what this paper claims.

5.1 Quantitative Tests

The framework makes specific numerical predictions. Each can be compared against observation. The "Paper" column indicates where each test originates:

Test	Prediction	Observed	Match	Paper
Tile convergence	Routes match	2.818 / 2.82 fm	0.21%	III (this)
Coupling ratio	$k^2 = 16$	15.93	0.41%	III (this)
$\sqrt{k/\pi}$ ratio	1.128	1.130	0.1%	III (this)
H from MOND	78.90 km/s/Mpc	(see note)	—	III (this)*
H_{measured}	69.92 km/s/Mpc	70.0 (GW170817)	0.11%	IV

Table 4: Quantitative tests from this paper and Paper IV. *Paper III predicted $H = 78.90$ km/s/Mpc from MOND; Paper IV proved this is H_{flat} and that MOND + topology yields $H_{\text{measured}} = 69.92$ (see §4.4).

Reading the table. The first three tests are internal consistency checks: does the topology hold? The fourth is this paper's prediction from MOND. The fifth is Paper IV's result, included for completeness. All matches are sub-percent.

5.2 Falsification Criteria

A scientific framework must specify what would disprove it. The following criteria would falsify this paper's claims:

> **Falsification Criteria**
>
> The framework would be **falsified** if:
>
> - Coupling ratio falls outside 16 ± 1 (currently: 15.93)
> - Tile size routes diverge by more than 1% (currently: 0.21%)
> - $\sqrt{k/\pi}$ ratio fails by more than 1% (currently: 0.1%)
> - Data requires $k \neq 4$

Why $k = 4$ is not adjustable. The value $k = 4$ comes from the simplex theorem: the minimum convex polytope enclosing volume in 3D has $d + 1 = 4$ faces. This is topology, not a fit parameter. If observations required $k = 5$, the framework would be falsified with no parameter to adjust.

What this paper does NOT claim:

- We do not derive r_e from pure topology (the scale emerges from convergence)
- We do not predict H_{measured} (Paper IV does this)
- We do not prove MOND is "correct" (we explain why a_0 has its value)
- We do not derive m_e from first principles (it is empirical input)

These boundaries matter. Overstating claims invites justified criticism. This paper's contribution is specific: deriving the tile size and identifying the $\sqrt{k/\pi}$ ratio. Paper IV extends these results to cosmological predictions.

6. Conclusion

This paper asked a specific question and provided a specific answer.

The question: Papers I and II established that gravity has $k = 4$ tetrahedral topology. But topology is dimensionless. What sets the tile SIZE?

The answer: Two independent routes converge on the same scale:

Route	Formula	Result
1 (Electromagnetic)	$r_e = e^2/(4\pi\varepsilon_0 m_e c^2)$	2.818 fm
2 (Gravitational)	$r_e = k\sqrt{G m_e/a_0}$	2.82 fm
Convergence		**0.21%**

Route 1 uses electromagnetism and special relativity. Route 2 uses gravity, cosmology, and topology. They share only the electron mass. Yet they agree to four significant figures. This convergence is unique to the electron.

The prediction: From the MOND equation $a_0 = cH/(2\pi)$, this paper predicts $H = 78.90$ km/s/Mpc. The ratio between cosmological and topological derivations of a_0 equals $\sqrt{k/\pi} = 1.128$ to 0.1%, connecting flat-space MOND to cosmological observables.

Subsequent development. Paper IV proved that MOND projected through $k = 4$ tetrad topology yields two distinct predictions: $H_{\text{flat}} = 78.90$ km/s/Mpc (MOND in flat space) and $H_{\text{measured}} = 69.92$ km/s/Mpc (MOND through topology). Both are valid for their respective contexts. Paper IV established that cosmological observations measure through the topology, so H_{measured} is what observers see, matching GW170817 to 0.11%.

Paper III Summary

Derived:	$r_e = 2.82$ fm (two routes, 0.21% convergence)
Identified:	$\sqrt{k/\pi} = 1.128$ (0.1% match to observed ratio)
Predicted:	$H = 78.90$ km/s/Mpc (from MOND equation)
Paper IV:	$H_{\text{measured}} = 69.92$ km/s/Mpc (0.11% match)

The broader implication. The unit of gravity is not just a scale; it implies field topology. The $k = 4$ tetrahedral structure that determines the tile size also determines how measurements project from local frames to cosmological boundaries. This paper identifies the tile size. Paper IV develops its implications for the Hubble constant.

Acknowledgments

The author thanks Mordehai Milgrom, whose identification of the acceleration scale a_0 in 1983 opened the research program that led to these insights. The observation that $a_0 \approx cH$ was not coincidental has been confirmed by the framework developed here.

Revision History

- **Version 5 (January 2026):** Complete rewrite for pedagogical clarity using the DFP Writing Framework. All sections restructured for improved readability: "why before how" exposition, clearer transitions, and better motivation of each result. Section 2 renamed "From Shape to Size" with improved tetrad formalism explanation. Section 4 renamed "The Topological Connection." Terminology updated throughout: "geometric" replaced with "topological" to align with standard tetrad literature (except where "geometry" correctly refers to the metric side of the tetrad equation). All equations unchanged. Two Hubble predictions now clearly distinguished: $H_{\text{flat}} = 78.90$ km/s/Mpc (MOND in flat space, this paper) and $H_{\text{measured}} = 69.92$ km/s/Mpc (MOND through topology, Paper IV). Both are valid predictions for their respective contexts. Historical Note (§4.4) documents this distinction. Previous versions available at Zenodo.
- **Version 4 (January 2026):** Pedagogical enhancements, learning objectives boxes, Paper V cross-references.
- **Version 3 (January 2026):** Initial public version.

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